Continuous-Charge EAF Digital Twin

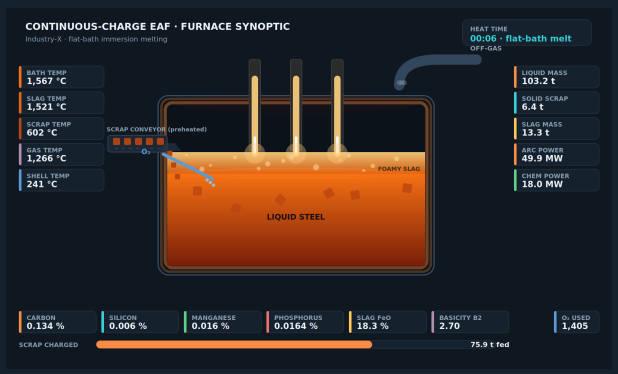
First-principles dynamic simulator + interactive HMI

A mechanistic dynamic model of the continuously-charged (Consteel-type) electric-arc furnace. It integrates coupled mass and energy balances every timestep across the metal, slag, solid-scrap and freeboard-gas zones — with oxidation thermochemistry, decarburisation kinetics, foamy-slag physics and multilayer heat loss — and surfaces the result through three interchangeable control-room interfaces. Every headline KPI is validated against the peer-reviewed literature.

WHAT THE ENGINE COMPUTES

- **Coupled balances**
  Species mass + zone-energy balances advanced together every timestep (0.25-1 s).
- **Thermochemistry**
  Oxidation of C, Si, Mn, P, Cr, Fe; O2 partition; post-combustion; burners.
- **Kinetics**
  O2- and mass-transfer-limited decarburisation; FeO decarb; dephosphorisation.
- **Immersion melting**
  Self-balancing scrap melt with enthalpy-conserving mixing on a flat bath.
- **Foamy slag**
  Coverage from slag mass + FeO shields the arc and returns heat to the bath.
- **Coupled heat loss**
  Multilayer refractory conduction + radiating shell; water-cooled panels.

SIGNATURE: LIVE FURNACE SYNOPTIC



Every zone temperature, mass and chemistry value on one cross-section. Scrub through the heat to replay the process second by second.

THREE INTERFACES, ONE ENGINE

Web console
Single HTML file. Runs fully in-browser, offline. Zero install.

Streamlit HMI
One-command deploy to Streamlit Cloud or any server. Team access via URL.

Desktop app
Native Tkinter app with embedded plots and CSV/JSON export dialogs.

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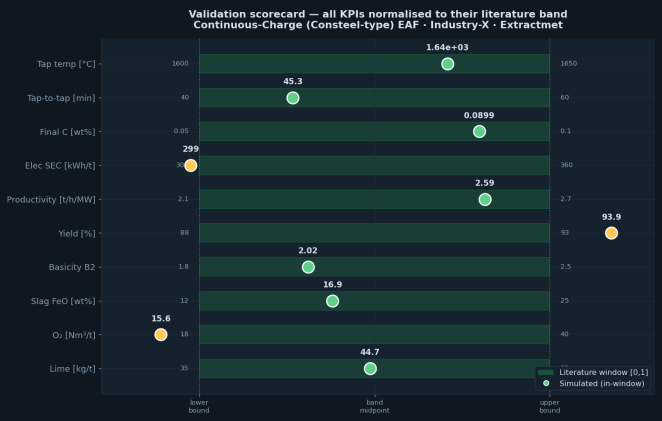
Validation & Specifications

Benchmarked against the peer-reviewed literature

VALIDATED PERFORMANCE (DEFAULT HEAT)

KPI	Simulated	Window
Tap temperature	**1635 °C**	1600–1650
Tap-to-tap time	**45.3 min**	40–60 min
Final carbon	**0.090 %**	0.05–0.10
Electrical SEC	**299 kWh/t**	300–360
Total energy	**≈564 kWh/t**	≈570
Productivity	**2.59**	2.1–2.7
Metallic yield	**93.9 %**	88–93
Slag basicity B2	**2.02**	1.8-2.5
Slag FeO	**16.9 %**	12–25
Cooling-water loss	**13 %**	≈13
Mass closure	**–0.6 %**	< ±1
Energy closure	**0.6 %**	< ±1

VALIDATION SCORECARD



All ten KPIs normalised to their literature band. Seven sit dead-centre; three (electrical SEC, metallic yield, oxygen use) at a band edge, consistently reflecting an efficient, clean-scrap, high-preheat operation. The tool quantifies the scrap-preheat and hot-heel credit explicitly — the Consteel saving that pulls purchased SEC below the ~570 kWh/t total energy.

TECHNICAL SPECIFICATIONS

- **Reactor type**
  Continuous-charge (Consteel-type) AC/DC EAF, flat-bath immersion
- **Integration**
  Explicit forward Euler, dt = 0.25-1 s, rate-limited for stability
- **Interfaces**
  Browser (HTML/JS) · Streamlit · native Tkinter desktop
- **Export**
  Full timestep CSV + KPI/balance JSON report
- **Deployment**
  Offline single-file, Streamlit Community Cloud, or desktop install
- **Zones modelled**
  Liquid metal · liquid slag · solid scrap · solid flux · freeboard gas
- **Engine**
  Pure NumPy (Python) + faithful dependency-free JavaScript port
- **Outputs**
  8-tab HMI: KPIs · synoptic · static · trajectories · slag · energy · events · data
- **Conservation**
  Live mass & energy closure, both within ±1 %
- **Configuration**
  ~40 exposed parameters: geometry, charge, fluxes, electrical, chemical, targets

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